



L8.2: What is an orthonormal basis?



Transcript Language

English



Hello, and welcome to the Maths 2 component of the online B.Sc. program on data science

and programming. In this video, we are going to talk about what is an orthonormal basis.

So, let us just recall first that we have defined what is an orthogonal set.

So, an orthogonal set of vectors v_1, v_2, v_k in an inner product space is a set of vectors

whose elements are mutually orthogonal. That means, if you the inner product v_i, v_j ,



where i is not equal to j , then this is 0. So, we checked in the previous video that an

orthogonal set of vectors is linearly independent. And that is why if you take a maximal orthogonal

set of vectors, then it is a maximal linearly independent set. And we have seen before that a

maximal linearly independent set is a basis. So, this is one way of getting a basis for an inner

product space. So, this is somewhat special. If you have an inner product on your vector space,

this is a slightly enhanced way of getting a basis. So,

such a basis was called an orthogonal basis. So, what is an orthonormal set? So, we are going

to use these two terms, an orthonormal set and a basis and put them together

and we will get an orthogonal, orthonormal basis. So, an orthonormal set of vectors

in an inner product space is an orthogonal set of vectors. So, that means the inner product of

v_i and v_j is 0 for all i not equal to j , such that the norm of each vector of the set is 1.

So, let us recall that, if you have an inner product, it automatically gives you a norm that is

if you have a vector v and you take the inner product of v, v , then that is a positive number,

non-negative number and if you take its square root, that gives you the norm. So,

that is defined as the norm of the vector. So, explicitly, what this means is if you have

a set v_1, v_2, v_k then S is an orthonormal set of vectors if the inner product of v_i and v_j is

0 for i and j in 1 through k and i is not equal to j , and the norm of v_i is 1. So, just to explicitly

say that norm of v_i is 1 is the same as saying that inner product of v_i, v_i is equal to 1.

So, here, of course, we do not take square root because a square root of 1 is 1. So,

we know that if the norm is 1, this is the same as saying the inner product is 1.

So, as an example, let us consider $\mathbb{R}^4$ with the usual inner product, that is the dot product. So,

then the set $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, 0, \frac{2}{\sqrt{42}}, \frac{1}{\sqrt{42}}, \frac{1}{\sqrt{42}}$

$\frac{6}{\sqrt{42}}, \frac{2}{3}, 0, \frac{2}{3}, -\frac{1}{3}$ is an orthonormal set of vectors.

So, this example, we have sort of seen a similar example before.

And the idea is here we have made them all into a norm, we have made all of them

to have norm 1. So, if you take the norm, so norm of the first vector is,

so this is the usual inner product, so the norm is just given by taking each component, squaring it,

and adding it up. So, that gives us one-third plus one-third plus one-third, which is 1.

So, similarly here, if you take the norm, that is going to give you 4 by 42 plus

1 by 42 plus 1 by 42 plus 36 by 42, which is indeed 1. And similarly, over here,

the norm is going to be given by two-third square plus two-third square plus minus one-third square,

that is 4 by 9 plus 4 by 9 plus 1 by 9, which is 1. So, all these have norm 1. And

you can check that the inner product is 0 if you take two different vectors.

So, for that, since all of them have the same denominator, you can ignore the denominator and

take the inner product and you can see clearly that the norm is indeed 0, sorry, the inner

product is indeed 0. So, I hope it is clear what is an orthonormal set. It is just an orthogonal

set with the additional property that the norms of each of the vectors in that set is 1.

So, now, what is an orthonormal basis? So, we know what is an orthonormal set. So, now,

an orthonormal set which forms a basis is an orthonormal basis. So,

this is similar to what we saw for the orthogonal basis, namely, an orthogonal basis was one where

it was an orthogonal set and it was a basis. So, here an orthonormal basis is one where it is

a basis and it is an orthonormal set.

So, equivalently an orthonormal basis is an orthogonal basis where the norm of each

vector is 1. So, because an orthogonal set is, orthonormal set is nothing but

an orthogonal set where each vector has norm 1 and orthonormal basis in particular is an orthogonal

basis where each vector has norm 1. So, equivalently, an orthonormal basis is

a maximal orthonormal set. So, just to make it clear what we mean by that, that means this set

is an orthonormal set and there is no set which is bigger than this, which properly contains this set

and which is also an orthonormal set. So, this is the largest possible orthonormal set you can

get. You cannot expand this set further and retain the property of being orthonormal.

So, here is an example. The standard basis with respect to the usual inner product

forms an orthonormal basis, maybe let us check that quickly before going ahead.

So, I have the vectors e_i . So, we already know that if you take e_i, e_j , then this is

and i is not equal to j , then this inner product is going to be,

so if you work this out, 0 times 0 plus 0 times 0 all the way, so, there is a lot of 0s.

And then when you come to the i th place, you will get a 0, sorry, a 1 time 0.

And when you come to the j th place,

you will get 0 times 1, this is if i is larger than, sorry, less than j then you get this. If

i is larger than j then the 0 times 1 comes first and the 1 times 0 comes next,

either way the point is you get this to be 0. So, this inner product is 0 and the norm of e_i ,

so that is e_i, e_i , so, well, square root of this, so then if you do the same computation,

you get 0 times 0 plus bunch of 0s. And then in the i th place there is one in each component,

so 1 times 1 and then again 0s. So, this gives you, I should have square root of this,

so which is square root of 1. And of course, when we say square

root and we are talking about norms, we always take the positive square root,

so, this is again 1. So, this is an example of an orthonormal basis. It is an orthogonal basis.

Well, we already know it is a basis. It is an orthogonal set that is what we checked

over here. So, here we checked it is orthogonal and here we checked it is orthonormal.

Let us do another example. So, consider $\mathbb{R}^3$ with the usual inner product and the set

one-third $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, one-third $\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$, and one-third $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$,

then this set which we have called β is an orthonormal basis. So, let us check this.

So, let us look at the norms first. So, let us give these names. So, let us call this,

let us call the first vector v_1 . So, we have v_1 , v_2 and v_3 . So, the norm of v_1 is

square root of, so, typically, if you want to, if you want to check that something has norm 1, then

it is enough to check that its square is 1. So, instead of checking that norm v_1 is 1, I will

be checking norm of v_1 squared is 1. So, norm of v_1 squared is just the inner product of v_1 itself,

which we can compute easily. So, this is one-third $1, 2, 2$,

one-third $1, 2, 2$, and that is 1 by 9 times 1 times 1 plus 2 times 2 plus 2 times 2 which is

1 plus 4 plus 4 so 9 by 9 which is 1 . The same, the same thing is going to happen

if you take v_2 , norm of v_2 which is one-third minus 2 , minus $1, 2$, one-third minus 2 , minus 1 ,

2 . So, this is giving us 1 by 9 times minus 2 times minus 2 plus minus 1 times minus 1

plus 2 times 2 , which is 9 by 9 , which is 1 . Maybe I will leave norm v_3 square

to you. So, check that this is 1 . And then we are left with these three

inner products. So, let us check v_1, v_2 what is the inner product. So, we have one-third $1, 2, 2$,

and one-third minus 2, minus 1, 2. So, again, the tip here is, if you want to check that the inner

product is 0, then these constants you can remove out and check for the term inside, which if it

is better term, so in this case, for example, you have integers,

then you would rather check that. So, in this case, you get 1 by 9 times

1 times minus 2 plus 2 times minus 1 plus 2 times 2. So, that gives us minus 2 minus 2 plus 4. So,

that is 0. I will again encourage you to check the other terms. So, this shows that this is

an orthonormal basis. Why is it a basis? We have checked here that this is an orthonormal

set, but the reason it is a basis is because it is a maximal orthonormal set.

Meaning, this is a set of size 3 and you already know it is a linearly independent set, because

if it is an orthonormal set, it is in particular an orthogonal set, which we have checked is

linearly independent. So, this is a, so I should end this by saying once you finish this checking,

this will so, this part will show and along with v_3 will show that the norms are 1.

This part if you finish, so check also that v_1, v_3 is v_2, v_3

is 0, this will, altogether will show that it is orthogonal. And then

the cardinality of β is 3 and β is linearly independent,

because it is orthogonal and we know that dimension of $\mathbb{R}^3$ is 3.

So, that implies that β is an orthonormal basis,

so because the sizes match. So, you have a linearly independent set of size which

is equal to the dimension of the vector space that is why it is a basis. So, in particular,

it is an orthonormal basis in this case. So, we have, I hope this gives you a window

into how to check something is a basis. Sometimes you happen to not know an explicit basis, but you

know the dimension of the space in consideration, so you can check that something is a basis.

So, let us talk about obtaining orthonormal sets from orthogonal sets.

So, suppose you have inner product space, an inner product space and if γ is a set v_1, v_2, v_k ,

which is orthogonal, so it is an orthogonal set of vectors. So, then we can obtain an orthonormal

set of vectors which let us call it β from γ by the following procedure. You take each

of these vectors and divide it by its norm then we are claiming that this is an orthonormal set.

So, why is it an orthonormal set? So, the reason is because this is exactly what we did in the

previous example. Each of those vectors was divided by its norm which made it orthonormal

and the orthogonality was evident already from just the integer part. So, here also that is

what is happening. So, here we have that v_1, v_2 is, so let us say $v_i \cdot v_j$ is 0, so this will imply

that v_i by norm v_i , v_j by norm v_j is 0. And so, this is orthogonal. So, this new

set is orthogonal β . So, the only thing that is remaining to check

is what is a norm. So, if you take the norm of v_i by norm v_i , well, constants come out of the norm.

So, you get 1 by norm v_i times norm v_i , which is 1. So, that is why this is an orthonormal set.

So, let us consider $\mathbb{R}^2$ with the usual inner product and the orthogonal basis

$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$. So, we can make this into an orthonormal basis by looking at

the norms of each of these vectors and then dividing by the term. So, the norm of both of

these vectors turns out to be $\sqrt{10}$. So, if you do $1/\sqrt{10}$ times $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and then $1/\sqrt{10}$ times

$\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ this is an orthonormal basis for $\mathbb{R}^2$.

So, the fact that it is an orthogonal set is already because γ was orthogonal. The fact

that it is an orthonormal set is because we divided by the norms. And the fact that it is a basis,

well, γ was already a basis. So, from there it follows this is a basis or the fact that $\mathbb{R}^2$ is of

dimension 2 and this is an orthonormal set of size 2 so it must be linearly independent, so linearly

independent of size 2, which tells you it is a basis.

So, maybe let me just write that again here, since it got lost out. So, here I was just saying that

$\langle v_i, v_j \rangle = 0$ implies $\langle v_i, v_j \rangle / (\|v_i\| \|v_j\|) = 0$. So, this is $1 / (\|v_i\| \|v_j\|)$ times

the inner product of v_i, v_j , which is 0. So, from here, we get that the existing set is orthogonal,

sorry, the new set is orthogonal. And norm of v_i by norm v_i

we can take the norm out. So, this gives us 1.

So, that is what, that is why this statement holds true. And the example

tells you, if you do not understand what I have written here, tells you why these things work. So,

work out the example for yourself. So, why are orthonormal bases important?

This is a sort of the punch line of what we are doing. So, suppose γ is v_1, v_2, v_n and this is

an orthonormal basis of an inner product space V and suppose you have a vector v inside capital V .

Well, we already know because it is a basis that you can write little v as $c_1 v_1$ plus $c_2 v_2$

plus $c_n v_n$. Why is that, because remember that a basis is in particular a spanning set, a basis

is a spanning set, which means every vector is a linear combination of the basis vectors.

And in fact, for a basis, it is a unique linear combination. So, there are a unique c_1, c_2, c_n

such that $v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$. So, this is a general statement for any basis. Now,

what is the importance of it, being an inner product space and with an orthonormal basis?

So, how do we find $c_1, c_2, \dots, c_n$, for any basis this means writing

a system of linear equations and solving. So, that is how I typically solve for $c_1, c_2,$

c_n . You write your v and then you write your $v_1, v_2, \dots, v_n$ and then you solve these equations.

But since γ is orthonormal, we can use the inner product and compute c_i is v ,

inner product of v, v_i . So, how do I do that? Let me quickly show you why that is true.

So, let us compute what is inner product of v, v_i . So, if you compute what is inner product of v, v_i ,

I substitute $c_1 v_1 + c_2 v_2 + \dots + c_i v_i + \dots + c_n v_n$ and then inner product with respect with v_i , well,

we know that the inner product is a linear in each variable, which means that I can write this as

c_1 times inner product of v_1, v_i plus c_2 times inner product of v_2, v_i , all the way up to,

then we have plus c_i times inner product of v_i, v_i plus

the inner product of v and v_i . This is what we get.

But now, we know that this is an orthonormal basis. So, because it is an orthonormal basis,

first of all, it is orthogonal. So, other than the v_i, v_i term, every other term is going to be 0. So,

this term will remain and all these terms are 0. So, this is, I can just write this as c_i times

v_i, v_i . So, this, we have used here that it is orthogonal, but now we also know

it is orthonormal, because it is orthonormal this is, we can rewrite this as c_i times norm

v_i squared, and norm v_i we know is 1. So, this is just c_i . This is where we are

using the fact that we have an orthonormal basis. So, what did we get? We got that

the inner product of v and v_i is equal to c_i . So, this is a very easy way of saying what

are the coefficients which come into this linear combination, which gives you v . How do we find c_1 ,

c_2 , c_n , the answer is each c_i is take the inner product of v with v_i and that is your c_i .

Let us do an example. We saw this orthonormal basis for $\mathbb{R}^2$ earlier.

So, write 2,5 as a linear combination in terms of these basis vectors. So, if you take 2,5

let us write this as c_1 times $\frac{1}{\sqrt{10}}$ times $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ plus c_2 times $\frac{1}{\sqrt{10}}$ times $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$.

Well, we just saw, what is c_1 . c_1 is the inner product of v, v_1 .

So, here v_1 is $\frac{1}{\sqrt{10}}$ times $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$, which is, you can pull out the constant, so $\frac{1}{\sqrt{10}}$

and then the remaining things, take the inner product, so this is 2 times 1 plus 5 times 3.

So, what does that give us, $\frac{1}{\sqrt{10}}$ times

2 plus 15, so that is 17. And then we get c_2 , which is v, v_2 ,

so $\frac{1}{\sqrt{10}}$ times minus 3, 1 which gives us,

again, the $\frac{1}{\sqrt{10}}$ comes out, and we get 2 times minus 3 plus 5 times 1, which is minus

6 plus 5, so minus 1 by $\sqrt{10}$. So, this is what we obtained as the coefficients.

So, now, if we write 2,5, in terms of these vectors we get 2,5 is $\frac{17}{\sqrt{10}}$ times v_1

plus minus $\frac{1}{\sqrt{10}}$ times v_2 , so which if you write out completely $\frac{17}{\sqrt{10}}$ times $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ times

$\frac{1}{\sqrt{10}}$ times $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$, sorry, $\frac{1}{\sqrt{10}}$ plus or rather minus $\frac{1}{\sqrt{10}}$ times,

1 by root 10 times minus 3, 1. And you can check actually that this is indeed going to work.

If you want, we can do that quickly. This is 17 by 10 minus

1 by 10 minus 3 , 1 . And if you work this out, the first entry is 17 by 10 minus 3 by 10 . So,

17 by 10 plus 3 by 10 so 20 by 10 , which is giving you 2 . And then the second entry is 5 by 10 minus

1 by 10 , which is giving you 5 . So, this shows that, indeed, what the equation we wrote down is

correct. This would have been maybe slightly harder if we did not have this knowledge that

this is an orthonormal basis. So, I hope in this video you have

figured out what, I mean, you have understood the main point. The main point is that we define

something called an orthonormal basis, that is an orthogonal basis in which each vector has

norm 1 . And the crux of the video is that every vector which you can write as a linear

combination of the orthonormal basis, the coefficients of, in that linear combination

c_1, c_2, c_n are essentially are equal to the inner product of the vector v with the corresponding

basis vector v_i . That is the main

point. So, I guess that finishes this video. Thank you.